

# continuation

## Continuation Document

**Revision:** 46 **Last modified:** 2026-06-01T14:11:33Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10  
**Maintainer:** Operator + AI loop

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### §0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

### §1: Recently Completed Work (last 10 commits)

| Commit  | Message                                                                                                                |
|---------|------------------------------------------------------------------------------------------------------------------------|
| 4d0ac66 | fix(swim,leader,node): resolve Wave 5 deferred — real-dep integration now green                                        |
| 8184725 | Foundation wave 5: mark 5 items Completed (HXC-1087/1089/1091/1092/1094); 1086/1088/1093 stay Queued (integration WIP) |
| 2f27ac9 | Foundation wave 5: 5 control-plane streams (gateway/lock/wireguard/node-registry proven; leader+swim+node-watch WIP)   |
| 8f570cb | fix(pkg/infra): isolate VMSpawn integration tests against deterministic vm-1 collision                                 |
| 23668fa | Foundation wave 4 + parity: mark 10 items Completed (HXC-1014/1075-1083)                                               |
| 44db918 | Foundation wave 4 + cross-platform parity: 9 real-infra items + macOS GPU/resources + real VMSSH — integration-proven  |
| 366a436 | CLAUDE-2: Cross-Platform Parity Guarantee — no Linux-only feature may fall back to a mock/stub on macOS                |
| f6209f6 | Foundation wave 3b: mark 7 items Completed (HXC-1010/1011/1012/1013/1015/1020/1028)                                    |
| 85ee85e |                                                                                                                        |

| Commit  | Message                                                                                                    |
|---------|------------------------------------------------------------------------------------------------------------|
|         | Foundation wave 3b: real-infra impl (infra orchestrator / podman builder / vault api) — integration-proven |
| 909d107 | Disable all active CI per operator directive: park race.yml in disabled/                                   |

## §2: Environment Snapshot

| Property  | Value                |
|-----------|----------------------|
| Branch    | main                 |
| Commit    | 4d0ac66              |
| Timestamp | 2026-06-01T14:11:33Z |

## §3: Active Work

| HXC     | Title                                                             | Status |
|---------|-------------------------------------------------------------------|--------|
| HXC-922 | Node service, advisory locks, chaos tests, docs                   | ✓ Done |
| HXC-921 | Web UI, K8s/Helm, integration, E2E, benchmarks                    | ✓ Done |
| HXC-920 | Health, security, build, wireguard services                       | ✓ Done |
| HXC-919 | htmux, messaging, GPU, WASM, storage, build manifest              | ✓ Done |
| HXC-918 | Wire scheduler/session gRPC to real backends, etcd discovery      | ✓ Done |
| HXC-916 | Stub expansion (config, retry, ratelimit, validator, build cache) | ✓ Done |
| HXC-914 | LLM, policy, setup, distributed lock                              | ✓ Done |
| HXC-913 | Health, security, ClassAds parser                                 | ✓ Done |
| HXC-912 | Gateway, helixd, helix-agent                                      | ✓ Done |
| HXC-911 | Scheduler, Session gRPC services                                  | ✓ Done |
| HXC-910 | Console, security, testing infra                                  | ✓ Done |

## §4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation
4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

## §5: Known Issues / Blockers

- None

## §6: Quick Commands

*# Run all tests (requires GOWORK due to workspace modules)*

```
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

*# Build all binaries*

```
GOWORK=$(pwd)/go.work go build ./cmd/...
```

*# Build web UI*

```
cd web && npm run build
```

*# Render Helm chart*

```
helm template helix-cluster deploy/helm/
```

*# Run benchmarks*

```
go test -bench=. ./test/benchmark
```

*# Run chaos tests*

```
go test -race -count=1 ./test/chaos
```